

SDG 14 • LIFE BELOW WATER

BlueGuard AI

Smart Ocean & Coastal Pollution Monitoring System

AI + IoT + Community Monitoring for Cleaner Oceans

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Date: 8/10/2025



1. Problem Statement

Why Life Below Water needs smarter monitoring



Plastic & Waste

Plastic and waste enter rivers, beaches and oceans, damaging marine ecosystems.



Late Detection

Pollution incidents and illegal dumping may be detected only after serious damage occurs.



Marine Life Threat

Fish, turtles, coral and other organisms are affected by polluted water and habitats.



Limited Monitoring

Manual and periodic testing cannot provide continuous coverage.



Need for Action

Communities need timely information to respond to pollution hotspots.

2. Proposed Solution

BlueGuard AI creates an early-warning network for cleaner oceans

IoT Sensors

Monitor pH, dissolved oxygen, turbidity, temperature and location.

AI Analysis

Analyze sensor and image data to identify pollution risks.

Camera / Drone

Detect visible waste and map pollution hotspots.

Live Dashboard

Show readings, alerts, locations and history.

Community Reporting

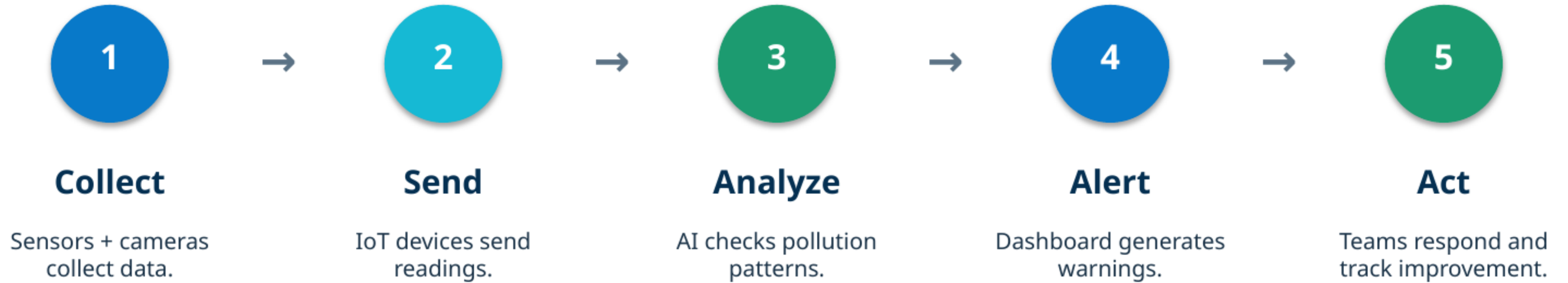
Collect local reports, photos and locations.

Cleanup Support

Use hotspot data to prioritize cleanup and prevention.

3. How the System Works

Simple end-to-end workflow



Early detection → faster response → healthier marine ecosystems

4. Key Features

Core features of the BlueGuard AI platform

Real-time monitoring

Continuous water-quality readings and system status.

AI pollution detection

Identifies abnormal readings and visual pollution indicators.

Pollution hotspot map

Shows high-risk areas for targeted action.

Alerts & notifications

Provides early warnings for communities and authorities.

Community reporting

Collects local reports, photos and locations.

Impact tracking

Measures waste removed, response time and hotspot reduction.

5. Lean Canvas Summary

Business/project model for BlueGuard AI

Problem

Plastic pollution
Late detection
Limited monitoring

Solution

IoT sensors
AI analysis
Camera/drone monitoring

Value Proposition

AI-powered early warning that helps protect marine life.

Customers

Coastal communities
Municipal bodies
NGOs
Fisheries

Channels

Web/mobile app
Colleges
NGOs
Social media

Key Metrics

Alerts detected
Response time
Water readings
Waste removed

Unfair Advantage

Local pollution dataset
Trained AI models
Partnerships

Costs

Sensors & hardware
Cloud hosting
AI/software
Maintenance

Revenue / Funding

Government contracts
NGO projects
CSR sponsorships
Grants

6. Expected Impact

How the project supports SDG 14 – Life Below Water



Cleaner water

Identify pollution before it spreads.



Protected marine life

Reduce risks to aquatic ecosystems.



Faster response

Give communities early, actionable alerts.



Better decisions

Use data to plan cleanup and prevention.

SDG 14: Conserve and sustainably use the oceans, seas and marine resources.

7. Website & Future Scope

Project demonstration and possible improvements

Project Website

Responsive website presents the problem, solution, features, impact and monitoring concept.

Future Scope

Connect real sensors; train AI with local datasets; add live maps and notifications; integrate NGO/government workflows.

GitHub Repository

github.com/sakshinayakawdi172-dotcom/BlueGuard-AI

Prototype to Product

Start with a low-cost coastal pilot, validate alerts, then scale to more locations.

Conclusion

BlueGuard AI combines AI, IoT and community participation to detect pollution early, support faster action and help protect life below water.

Cleaner Oceans • Healthier Marine Life • Sustainable Future

Thank You! 

Sakshi Nayakawdi | SDG 14 – Life Below Water